A Clinical Workflow Integration Layer for Tuberculosis Chest X-ray Screening

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***Abstract*—Tuberculosis (TB) causes more deaths than any other infectious disease worldwide. However, high-burden areas are often short-staffed on radiologists to interpret chest X-rays for TB screening. Hence, computer-aided detection (CAD) systems for TB screening are in wide use. While TB detection models are highly accurate, several practical challenges emerge when integrating these systems into clinical workflows. Commercial TB CAD systems often do not account for X-ray image quality, post-deployment monitoring, and explanation of findings in natural language for interpreting radiologists. We present a prototype of a clinical workflow integration layer to address these issues. Its lead contribution is a large language model explanation module that describes model findings in natural language for radiologists, a feature not yet present in the commercial products we reviewed. Supporting modules add an input quality gate to assess X-ray image quality, a structured evidence interface to present findings, and a post-deployment monitoring dashboard. Alongside the workflow layer, we also develop and benchmark custom models for tuberculosis classification and localization. Our results show that ... (TODO).**

***Keywords*—Tuberculosis; Computer-Aided Detection; Chest X-ray; Clinical Workflow; Medical Imaging; Computer Vision**

I. Introduction

Tuberculosis (TB) causes more deaths than any other infectious disease worldwide [1]. It is caused by airborne *Mycobacterium tuberculosis* and primarily affects the lungs [3]. Low- and middle-income countries bear a disproportionate share of the burden, compounded by limited access to diagnostics, delayed presentation, and overburdened health infrastructure [2].

Chest X-ray (CXR) is the most widely used triage modality for TB screening [4]. If TB is suspected, bacteriological testing (sputum smear or molecular assays) is conducted to confirm TB [2]. However, high-burden countries are often short-staffed on radiologists to interpret CXRs, especially in primary care and rural units where volumes are highest [2]. The shortage has driven the development of computer-aided detection (CAD) systems that flag CXRs suggestive of TB [6].

Radiologists agree with AI findings on most cases, and AI assistance is found to improve radiologist performance. A reader study found that AI assistance raised physician AUC from 0.773 to 0.874 and enabled non-radiologists to match trained radiologists [10]. In real-world use, radiologists were in complete agreement with AI on 86.5% of cases, with only 0.5% of findings missed by the model [11].

A. Maturity of TB CAD Systems

TB CAD systems have matured into routine clinical use. The World Health Organization (WHO) endorsed CAD as an alternative to human readers in 2021 [6], and in 2025 approved six commercial products as meeting its TB screening performance standards [7]. Reviews report model accuracy approaching WHO targets across populations [9], [41], [42], indicating that the field has moved beyond accuracy toward integrating TB CAD into routine clinical workflows.

B. Gaps in Clinical Workflow Integration

Reviews and deployment studies consistently identify gaps between laboratory accuracy and clinical implementation. Only a small proportion of published CXR AI studies are prospective or clinically embedded, and workflow integration, local validation, and human factors remain under-studied [12], [9]. Field deployments report similar issues: an India deployment identified barriers in workflow, connectivity, and interpretation [38], a Vietnam deployment showed only 47.3% agreement between physician and CAD interpretations at three months [39], and clinicians across countries valued CAD most as a complement rather than a replacement [40]. Usability testing further showed that users wanted clearer explanations of model findings [13]. Three concrete gaps emerge:

CXR image quality varies in real-world use. CAD systems are trained on curated datasets, but routine hospital images may be poorly positioned, clipped, underexposed, or photographed from analog film [29]. Poor input quality degrades performance, yet most systems do not expose a user-facing quality assessment feature before inference [12], [14].

Heatmaps for explaining model findings lack clinical utility. All commercial CAD products in Table I provide heatmaps of regions that influenced the model's prediction. However, radiologist studies report that these are less intuitive than discrete localized findings [44], often overlap poorly with expert-identified regions, and do not convey the contextual reasoning used during diagnosis [30]. Textual descriptions and case-based reasoning have been proposed as more clinically meaningful alternatives [31], [15].

Post-deployment monitoring is often lacking. Deployed performance may shift due to population, equipment, or software changes [55]. The U.S. Food and Drug Administration has identified these as risks to AI-enabled medical devices and initiated research on proactive post-market monitoring [34], but in practice this remains an operational gap [12].

C. Related Work

Academic prototypes have explored vision-language models to explain findings in natural language. Siemens Healthineers, who also developed one of the commercial TB CAD

TABLE I. FEATURE COMPARISON WITH COMMERCIAL TB CAD PRODUCTS

Feature	**qXR** [22]	**CAD4TB+** [23]	**Lunit** [24]	**DrAid** [26]	**Genki** [27]	**InferRead** [28]	**Siemens** [25]	**Proposed**
Chest X-ray quality assessment				*n/a*	*n/a*		✓	✓
Monitoring and analytics dashboard			✓	*n/a*	*n/a*			✓
Heatmap/overlay of localization findings	✓	✓	✓	✓	✓	✓	✓	✓
Explanation of findings in natural language				*n/a*	*n/a*			✓

✓ marks a feature documented as present; blank marks a feature documented as absent or not offered; *n/a* marks features for which no documentation was found.
The monitoring/analytics dashboard column denotes a post-deployment dashboard for tracking model usage and performance over time.

systems compared in this paper, pair a combined classification-localization model [16] with a domain-adapted LLM to generate CXR findings [15], although this feature is not yet in their commercial product. General medical vision-language models such as LLaVA-Med [17] and CXR-LLaVA [32] demonstrate image-conditioned radiology text generation, and recent surveys catalogue the rapid growth of vision-language models for medical report generation [33]. However, these remain research prototypes rather than features in deployed systems or commercial products.

We compare against seven commercial CAD products: six WHO-approved (qXR v4.0 [22], CAD4TB v7 [23], Lunit INSIGHT CXR v3.1 [24], DrAid v2.4.4–6 [26], Genki Edge v3.4.2 [27], InferRead DR Chest v1.0.1.1 [28]) [7] plus Siemens AI-Rad Companion [25], which uniquely implements input quality assessment and has explored language-based explanation in research [15]. As Table I summarizes, all provide heatmap localization, but post-deployment monitoring is uncommon, input quality assessment is largely absent, and no product is known to explain findings in natural language.

D. Scope and Contribution

This paper presents a prototype clinical workflow integration layer for TB CAD that targets the gaps above. Its contributions are:

1) A **natural language explanation module** (lead contribution) that describes AI-detected findings and suspected abnormalities with clinical grounding.
2) An **input quality assessment module** that screens radiographs for acquisition defects before inference.
3) A **post-deployment monitoring dashboard** tracking system behavior and performance over time.

As a secondary contribution, we develop and benchmark custom models for classification (Section IV) and localization (Section V).

II. SYSTEM ARCHITECTURE

The system processes a CXR through a sequential pipeline described in Figure 1. A CXR (DICOM or standard image format) is ingested from the clinical imaging system or uploaded directly, then evaluated for acquisition quality before any diagnostic inference. Images passing quality screening are scored by the TB classification module. For cases flagged as TB, localization (object detection) is done to detect active and latent TB. Findings are passed to a module that generates a natural language description of the finding based in clinical context. Outputs render in a structured evidence panel, and each case is logged for post-deployment monitoring.

III. INPUT QUALITY ASSESSMENT

IV. TUBERCULOSIS CLASSIFICATION

The TB classification module flags each CXR as either healthy, sick non-TB, or TB as seen in Figure 1.

A. Dataset

Classification and localization models are trained on TBX11K [50], a public CXR dataset with image-level labels (*healthy*, *sick non-TB*, *TB*) and bounding-box-level labels (*active-TB*, *latent-TB*). We use a simplified redistribution[1] with 8,400 CXRs (3,800 healthy, 3,800 sick non-TB, 800 TB), with a 79%/21% train/validation split (6,600 and 1,800 images). TBX11K is selected since it is the largest public TB CXR dataset combining a TB-specific sample large enough to train from scratch, expert bounding-box annotations, a sick non-TB class, and a demographic and class composition that reflect real-world clinical settings [51].

The official TBX11K test set is withheld for an online challenge, so we evaluate on the public validation set. Our results are therefore not directly comparable to SymFormer [51] or other challenge entries on the private test set.

B. Data Preprocessing

Classification models are trained without preprocessing or data augmentation. CXRs unfit for inference are already filtered out by the input quality assessment module. CXRs are also acquired under a standardized posteroanterior protocol and are largely uniform in pose, so aggressive augmentation risks introducing distribution shift.

C. Model Architecture

We evaluate nine classifiers: six baselines (EfficientNet-B0 [62], MobileNetV3-Large [61], DenseNet-121 [60], ResNet-18, ResNet-50 [59], ConvNeXt-Tiny [63]) and three custom models (DraxNet, Drax-MobileNetV3-Large, FlipR).

1) DraxNet: DraxNet (17.0M parameters) modifies a ResNet-18 backbone so the fourth layer consists of two ResNet blocks, each with a custom mixer (*DraxBlock*) inserted between its two convolutions. All other components are unchanged. *DraxBlock* consists of a ConvNeXt local convolutional branch followed by an optional self-attention branch, fused via stochastic-depth residuals. DraxNet, short for Dynamic Residual Attention eXchange, reserves the heavier mixer for the final, lowest-resolution stage, adding attention capacity with fewer parameters than a full attention backbone.

[1] https://www.kaggle.com/datasets/vbookshelf/tbx11k-simplified

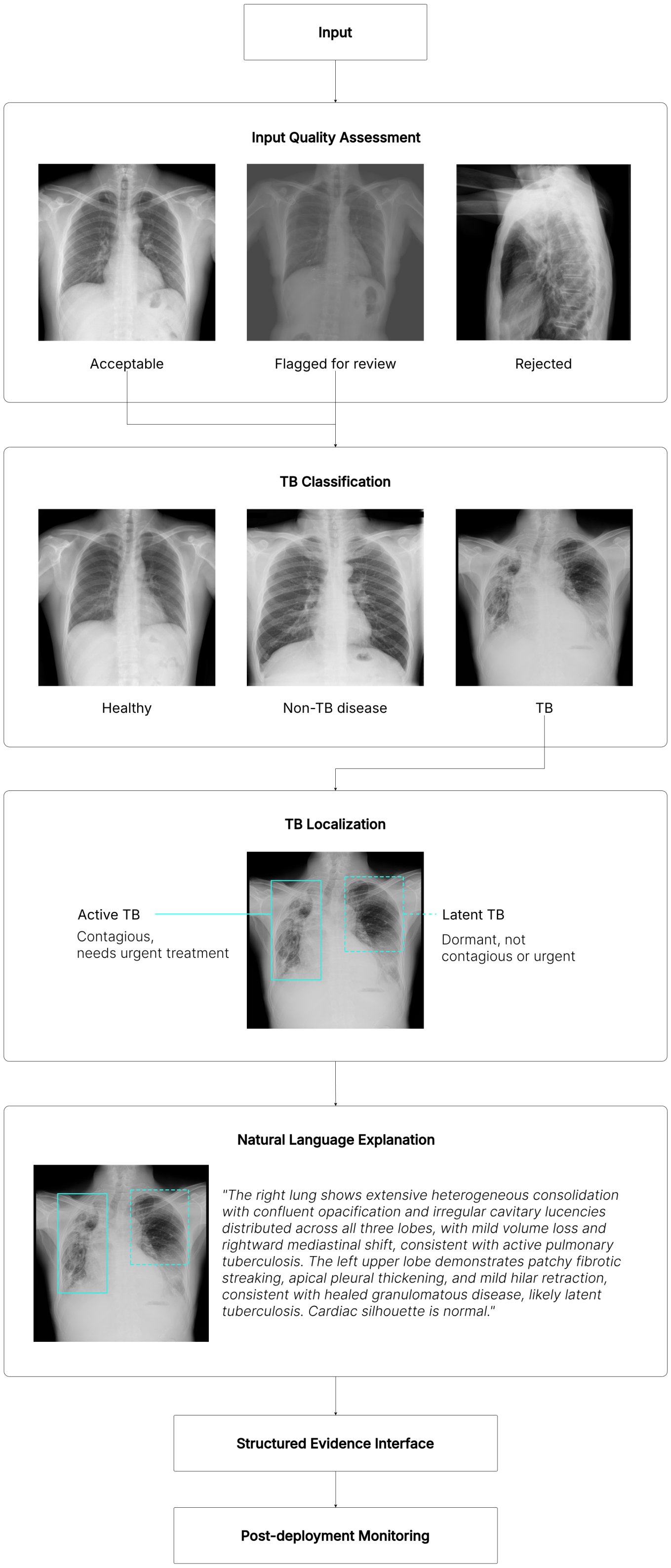

Fig. 1. Overview of the proposed TB CAD pipeline.

2) Drax-MobileNetV3-Large: Drax-MobileNetV3-Large (6.1M parameters) applies the DraxBlock to a parameter-efficient backbone. The MobileNetV3-Large feature extractor and head are unchanged, and a bottlenecked *DraxBlock* is inserted after the final feature stage. This tests whether the DraxBlock generalizes beyond a ResNet-style backbone.

3) FlipR: FlipR (2.8M parameters) inserts a *FlipR* block between layers of an otherwise unchanged ResNet-18. The *FlipR* block computes the difference between each feature map and its horizontally flipped counterpart, smoothed by average pooling to account for imperfect anatomical symmetry, then broadcasts a gated mask across all channels. The block is inspired by SymAttention [51], which observes that bilateral asymmetry in CXRs is a key indicator of TB. FlipR tests whether a lightweight symmetry gate can match heavier attention approaches.

D. Model Training

All models are trained from scratch with a fixed random seed, batch size 16, 512×512 input, 100 epochs, Adam at a fixed learning rate of 10^{-3}, no schedule, and unweighted cross-entropy loss. No hyperparameter tuning was done.

E. Results

Classification results are reported in Table II.

MobileNetV3-Large is the top classifier, while FlipR achieves comparable results with half the parameters. Given the small ($\pm 1\%$) differences, ordering among the top four (FlipR, EfficientNet, MobileNetV3-Large, Drax-MobileNetV3-Large) is likely due to stochasticity. They differ most on sensitivity ($\pm 2.5\%$), so we interpret MobileNetV3-Large (96.67% sensitivity) as the top model, since sensitivity is the clinically dominant metric in TB screening. FlipR reaches comparable results on other metrics at roughly half the size (2.8M vs 5.5M), supporting the SymFormer [51] hypothesis that bilateral asymmetry is highly indicative of TB.

All models meet WHO targets for TB triage. Sensitivity ranges from 92.50 to 96.67%, clearing the WHO target of 90% sensitivity. Specificity ranges from 97.98 to 99.91%, well above the WHO target of 70% specificity [8]. Sensitivity is more clinically important than specificity in screening, since a missed TB case risks untreated transmission while a false positive triggers only confirmatory testing. The top model, MobileNetV3-Large, misses only 3.3% of TB cases.

The high-specificity, lower-sensitivity profile is likely attributable to TBX11K rather than model architecture. Models in the literature consistently show the same pattern on TBX11K. The dataset is constructed to reward high specificity, since the common failure mode on other TB datasets is over-predicting TB and collapsing specificity to near-zero [51].

Lightweight models are preferred for TB screening on both performance and deployment grounds. Smaller models outperform larger ones across the board: MobileNetV3-Large (5.5M), EfficientNet-B0 (5.3M), and FlipR (2.8M) beat all larger models in accuracy and F1. Due to a modest TBX11K training set, a minority TB class, and no class-imbalance handling, larger networks appear to overfit the majority healthy and sick non-TB classes rather than learn discriminative TB features. TB screening also runs in resource-limited settings often without a dedicated GPU, so a lightweight model is preferable in deployment.

TABLE II. TUBERCULOSIS CLASSIFICATION RESULTS

Model	AUC_{TB}	Acc. (%)	Sens. (%)	Spec. (%)	Avg. Prec. (%)	Avg. Rec. (%)	Avg. F1 (%)	Params
FlipR	0.9984	98.97	94.17	**99.91**	**99.01**	97.70	98.34	**2.8M**
EfficientNet-B0	**0.9995**	98.89	95.00	99.65	98.29	97.87	98.07	5.3M
MobileNetV3-Large	**0.9995**	**99.21**	**96.67**	99.82	98.97	**98.54**	**98.75**	5.5M
Drax-MobileNetV3-Large	0.9989	98.81	95.00	99.65	98.23	97.81	98.02	6.1M
DenseNet-121	0.9975	98.57	92.50	99.56	97.81	96.97	97.38	8.0M
ResNet-18	0.9982	98.17	93.33	99.21	96.71	96.90	96.80	11.7M
DraxNet	0.9988	98.41	95.00	99.47	97.51	97.51	97.51	17.0M
ResNet-50	0.9978	97.70	92.50	99.47	96.95	96.33	96.63	25.6M
ConvNeXt-Tiny	0.9952	97.14	95.00	97.98	93.63	96.58	94.97	28.6M

AUC_{TB} is the area under the ROC curve for the TB class. Sensitivity is recall for the TB class. Specificity is recall for the non-TB class (healthy and sick non-TB combined). Precision, recall, and F1 are averaged across the three classes (healthy, sick non-TB, TB).

V. TUBERCULOSIS LOCALIZATION

A. Dataset

Localization uses the TBX11K bounding-box annotations, with each region labeled active TB or latent TB [50]. Active TB denotes ongoing, transmissible infection requiring treatment; latent TB denotes prior, inactive infection that is not contagious but may reactivate. As Figure 1 shows, active TB is more visually salient and latent TB is more subtle. A TB-labeled CXR may contain only active-TB lesions (924 images), only latent-TB lesions (212 images), or both (54 images), a composition consistent with real-world conditions.

B. Model Architecture

We evaluate YOLO26 and a custom variant in which the detector's default feature extractor is replaced by the DraxNet backbone used in classification. The detection head and training protocol are unchanged, isolating the backbone's effect on localization.

C. Model Training

Localization uses the same training configuration as the classification models (Section IV).

TABLE III. TUBERCULOSIS LOCALIZATION RESULTS

Model	Class	mAP_{50}	$mAP_{50\text{-}95}$
YOLO26	ATB	—	0.2110
YOLO26	LTB	—	**0.0410**
YOLO26 + DraxNet	ATB	—	**0.2748**
YOLO26 + DraxNet	LTB	—	0.0310

ATB = active TB; LTB = latent TB.

D. Results

Localization results per class are reported in Table III.

Active TB is more likely to be detected than latent TB. The very low $mAP_{50\text{-}95}$ for latent TB reflects the TBX11K class imbalance (active-TB regions outnumber latent-TB regions roughly 4:1) and the subtle appearance of latent-TB lesions. The TBX11K reference reports the same pattern across all detection methods, attributing it to imbalance [51], and it is consistent with human readers distinguishing latent TB less reliably [5]. The bias toward active TB also aligns with clinical triage priorities, since active TB requires urgent treatment.

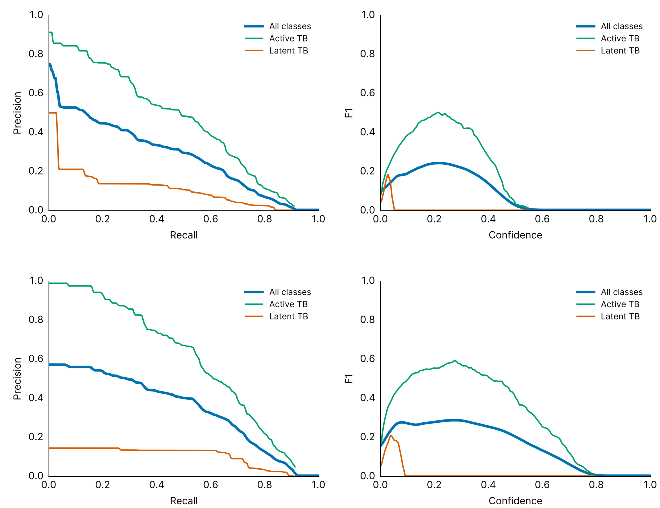

Fig. 2. Localization precision-recall (left) and F1-confidence (right) curves for YOLO26 (top) and YOLO26 with the DraxNet backbone (bottom). Both sustain high precision while recall saturates near 0.26, and the latent-TB curve collapses at low confidence, so localization functions as a high-precision rule-in cue rather than a standalone detector.

The DraxNet backbone specializes toward active TB. YOLO26 with the DraxNet backbone raises active-TB $mAP_{50\text{-}95}$ from 0.2110 to 0.2748 (+0.0638) while slightly reducing latent-TB (0.0410 to 0.0310). The gain is driven almost entirely by precision ($0.74 \to 0.82$). Recall is unchanged at roughly 0.26.

Recall, not precision, is the bottleneck. Both detectors localize confidently but find only about a quarter of the annotated TB regions. Figure 2 shows precision staying high while recall saturates near 0.26, and the latent-TB curve collapsing at low confidence for both detectors.

Localization is useful to confirm TB, not to rule it out. The high-precision, low-recall profile dictates clinical use: a flagged lesion is likely correct and warrants focused radiologist review, but the detector must not exclude TB since most annotated lesions are missed. Localization is therefore most reliable as supporting region-level evidence for the classifier and LLM explanation modules, rather than as a standalone detector.

E. Implementation Notes

Localization runs only on CXRs the classifier (Section IV) flags as TB. Non-TB CXRs skip localization, since any lesions found would contradict the classifier's decision.

VI. Multimodal Explanation

VII. Structured Evidence Interface

VIII. Post-Deployment Monitoring

IX. Discussion

A. Limitations

B. Future Work

X. Conclusion

Declaration on Generative AI

Generative AI tools were used to assist in drafting and preparing this manuscript. The conception of the research, methodology, and conclusions are the work of the authors.

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